

Contrastive Learning using Random Walk Laplacian

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Background on Self-Supervised Learning (SSL)

- Feasibility and cost of labeling
- SSL : a more "ecological" way of learning ?

- Representation learning
- Cross-domain / Out-of-domain generalization
- Zero-shot / few shot learning
- Multimodal learning

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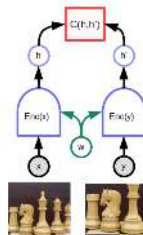
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Background on Self-Supervised Learning (SSL)

Self-prediction



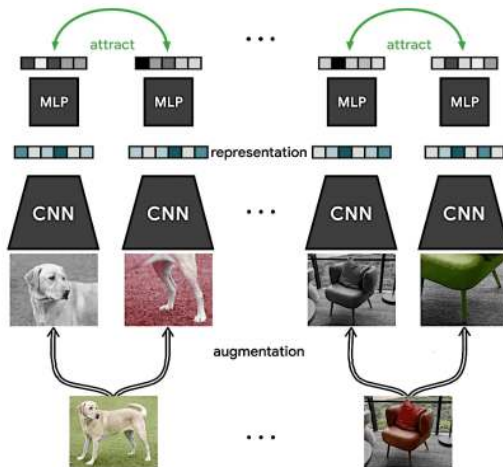
Contrastive Learning



<https://ai.facebook.com/blog/self-supervised-learning-the-dark-matter-of-intelligence/>

Successful SSL approaches

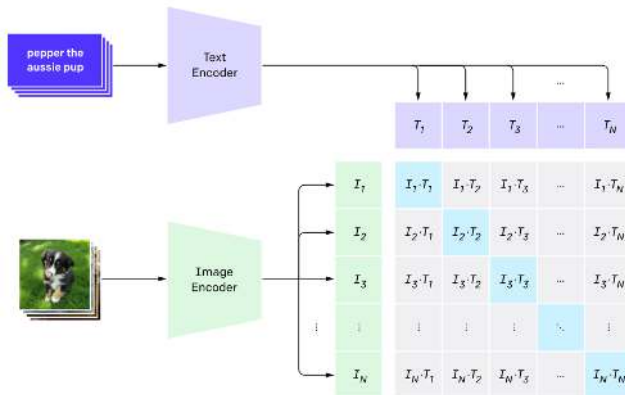
SimCLR



Chen, Ting, et al. A simple framework for contrastive learning of visual representations. ICML 2020.

Successful SSL approaches

Contrastive learning with text and image / audio



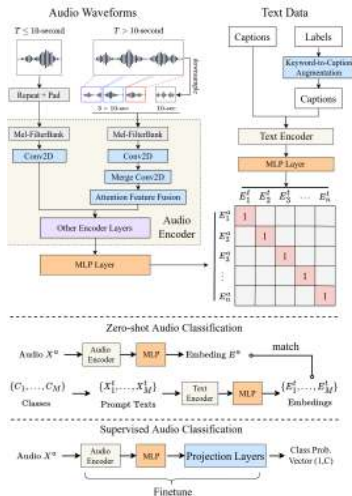
CLIP pre-trains an image encoder and a text encoder to predict which images were paired with which texts in our dataset. We then use this behavior to turn CLIP into a zero shot classifier. We convert all of a dataset's classes into captions such as "a photo of a dog" and predict the class of the caption CLIP estimates best pairs with a given image.

<https://openai.com/research/clip>

CLIP <https://arxiv.org/abs/2103.00020> Radford et al 2021

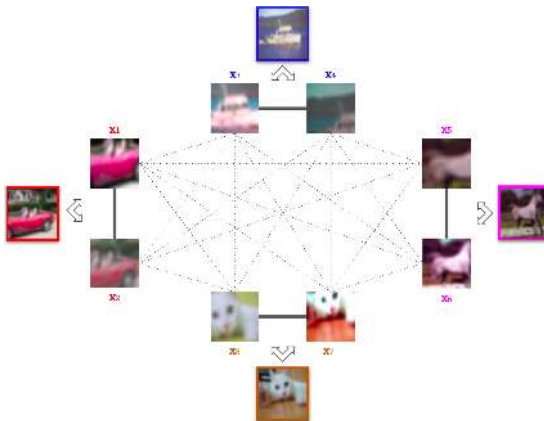
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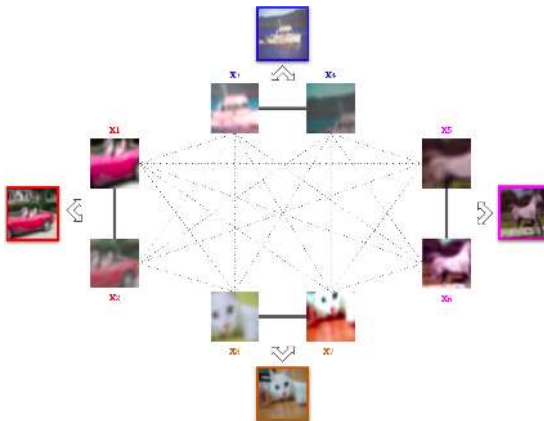
This work

A graph perspective on contrastive learning



This work

A graph perspective on contrastive learning



(UniVCL (Tang et al., 2022, ECCV) proposed a unifying perspective.)

Contrastive Learning using the RW Laplacian (CLRW)

- Weighted adjacency matrix $\mathbf{W}$ as :

$$\mathbf{W}_{i,j} = \begin{cases} \exp(f_{\theta}(\mathbf{x}_i)^{\top} f_{\theta}(\mathbf{x}_j)/\tau), & \text{if } i \neq j \\ 0, & \text{otherwise} \end{cases}, \quad (1)$$

where f_{θ} : encoder with parameters θ

- Degree $\mathbf{D} = \text{diag}\left(\left[\sum_{j=1}^N \mathbf{W}_{i,j}\right]_i\right)$
- Random Walk Laplacian $\mathbf{P} = \mathbf{D}^{-1}\mathbf{W}$.
- Loss function for examples i, j in a batch $\mathcal{B}$

$$\mathcal{L}(\mathbf{x}_i, \mathbf{x}_j) = \mathbf{1}_{[i \sim j]}(1 - \mathbf{P}_{i,j}) + \mathbf{1}_{[i \not\sim j]} \mathbf{P}_{i,j}, \quad (2)$$

$$\mathcal{L}^{\mathcal{B}} = \sum_{\substack{\mathbf{x}_i, \mathbf{x}_j \in \mathcal{B} \\ i \neq j}} \mathcal{L}(\mathbf{x}_i, \mathbf{x}_j). \quad (3)$$

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- SSL pretraining with a Resnet18 backbone on CIFAR10
- Two views in each batch ($N = 256$) using data augmentation
- In-domain (same dataset) linear Evaluation on 100 epochs
- Comparison with SimCLR

Method	Epochs	Accuracy
SimCLR (reproduced)	100	83.5 ± 0.20
SimCLR (reproduced)	800	91.0 ± 0.1
CLRW	100	83.7 ± 1.32
CLRW	800	91.1 ± 0.1

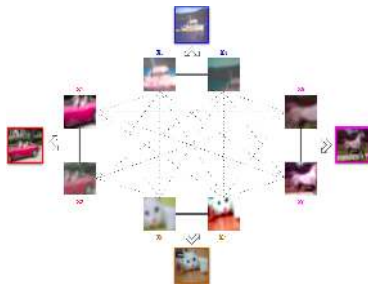
Discussion

Main findings

- CLRW obtains similar performances than SimCLR
- CLRW can be seen as a reformulation of SimCLR

Perspectives

- Exploit the graph structure using graph operators / filters
- Data augmentation in the latent space using the graph
- Generate more views and use self-loops



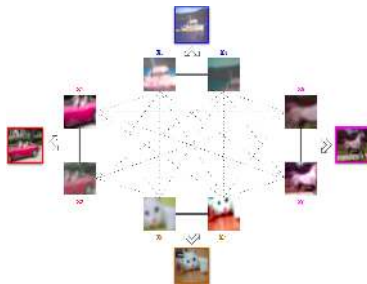
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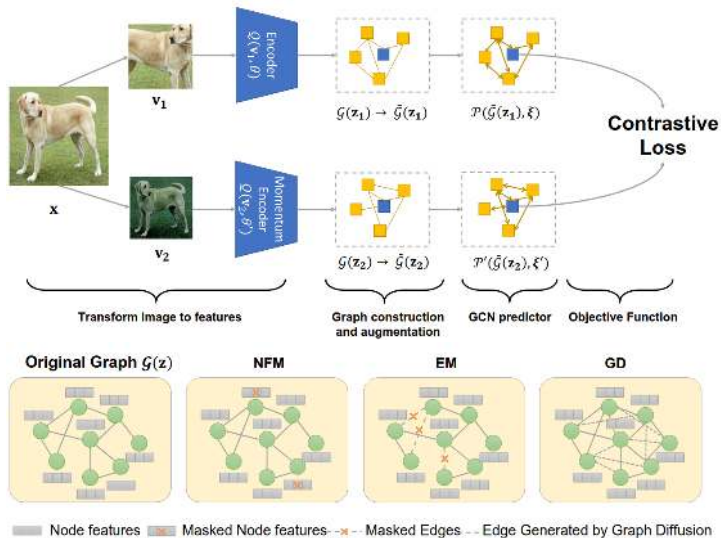
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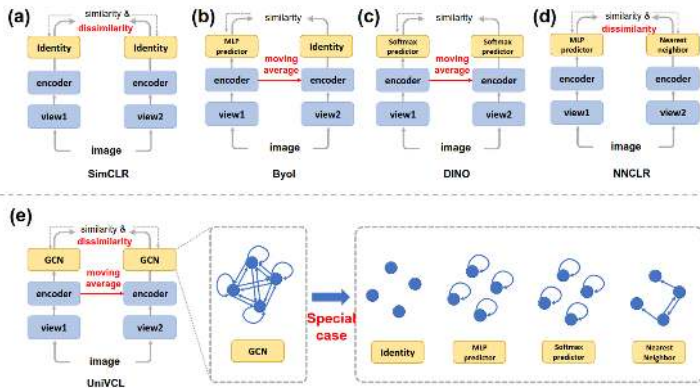
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(UniVCL (Tang et al., 2022, ECCV))

- Use labels or metadata to define edges (similar to Supervised Contrastive Learning)
- GSP operators / filters or other GNN architectures in the latent space
- Exploit class hierarchies to build edges
- Multimodal settings

Supervised Contrastive Learning <https://arxiv.org/abs/2004.11362>,
using metadata Moummad and Farrugia 2022 <https://arxiv.org/abs/2210.16192>